

Episode 13

Tech Talks South Tyrol



27.05.2026



PROGRESS **GROUP**



Andreas



Martin



PROGRESS **GROUP**

WELCOME



Global partner in the
precast industry



80 countries
with customer relations



750 employees
in the group



300 Mio. €
Turnover

COMPLETE PRECAST TECHNOLOGY WORLDWIDE

BUSINESS SEGMENTS

PROGRESS **GROUP**



BUILDING SYSTEMS

Innovative building systems with precast concrete elements



MECHANICAL ENGINEERING

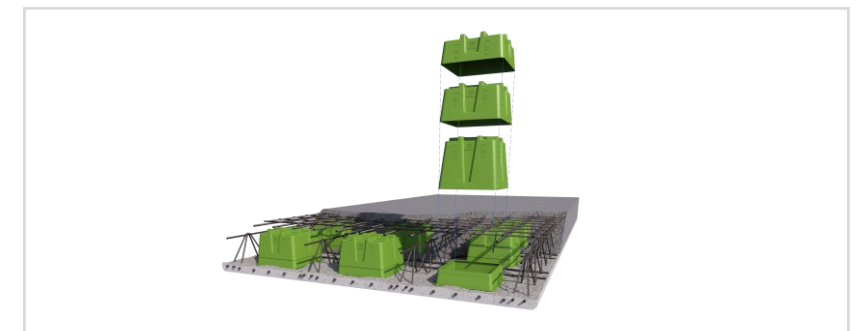
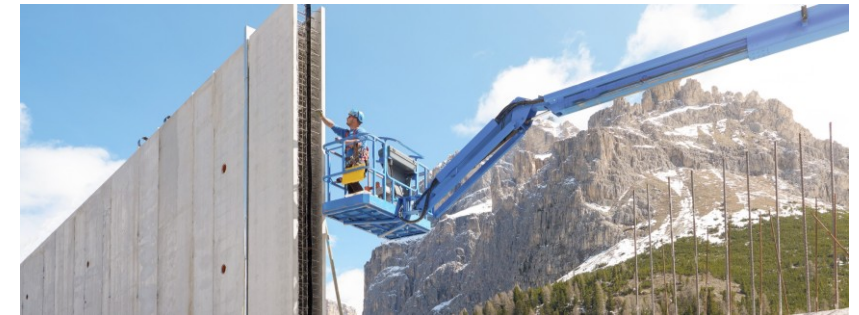
Innovative technology for the precast and rebar industry



SOFTWARE DEVELOPMENT

Innovative software solutions for the precast and rebar industry



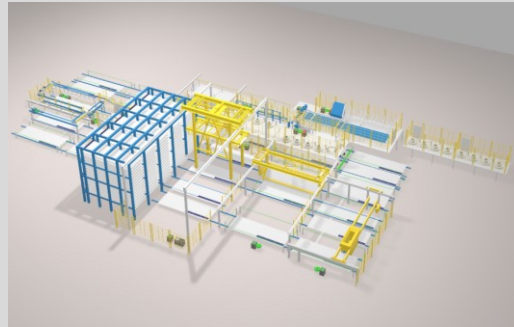


AUTOMATION



Flexible machines with high degree of automation

INDUSTRY 4.0



Digitization and networking of production systems

ROBOTICS

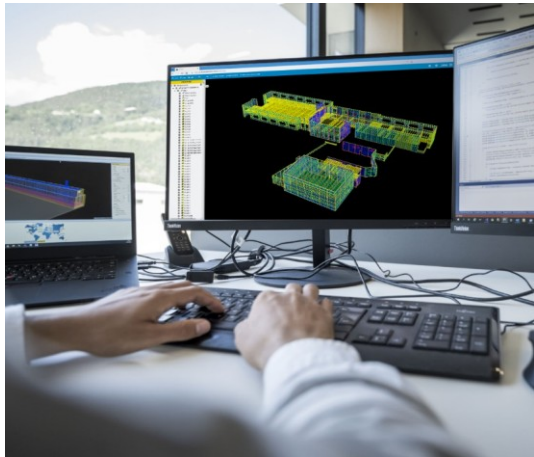


Modern robot technology



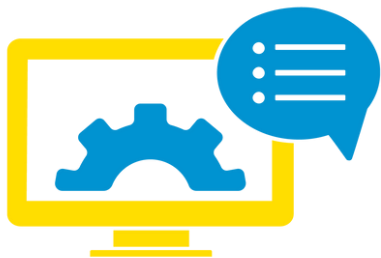
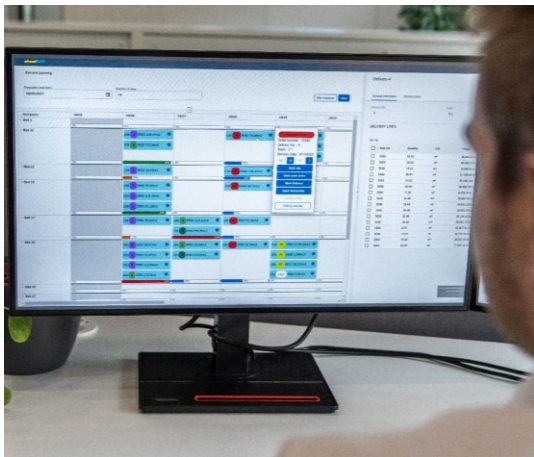
DESIGN

Solutions to support the BIM process



PLANNING

ERP for complete mapping of business processes



PRODUCTION

MES for planning, controlling and optimize production

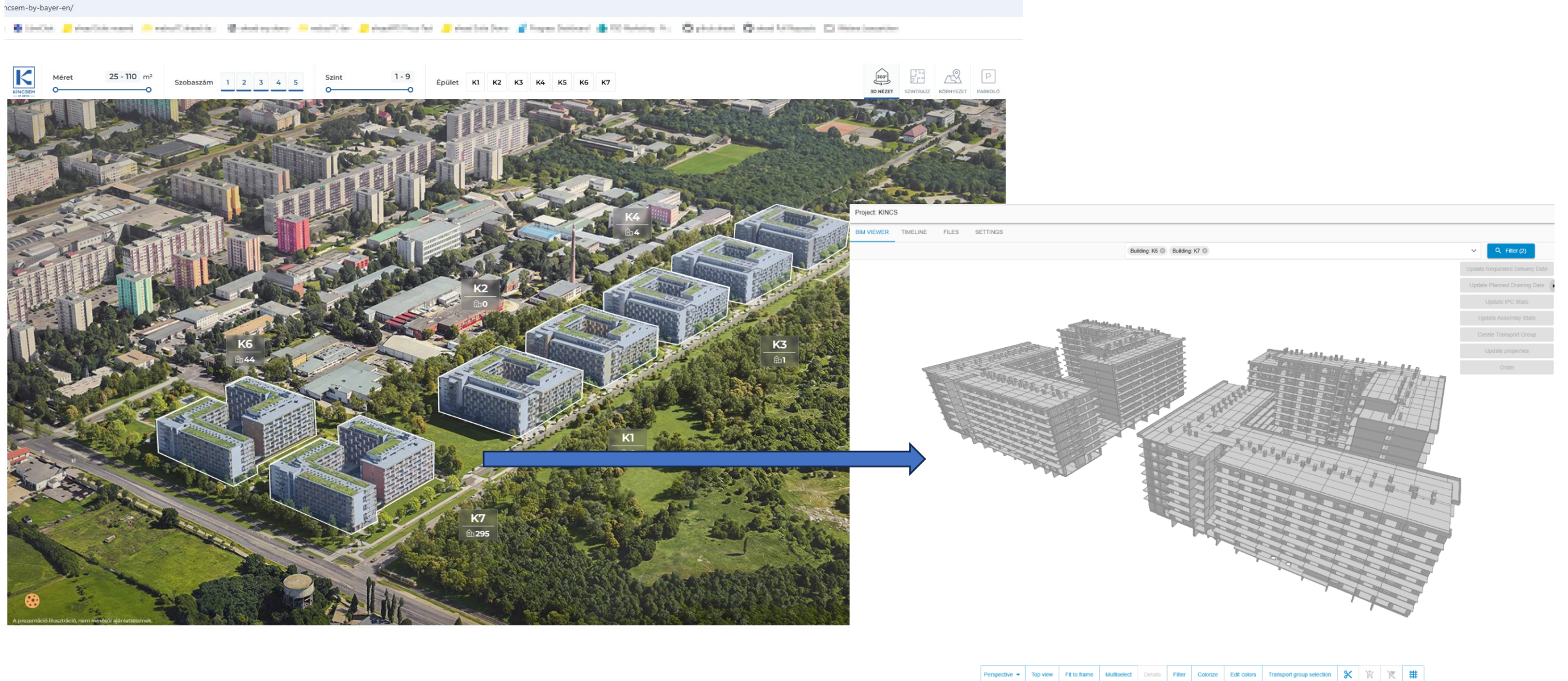


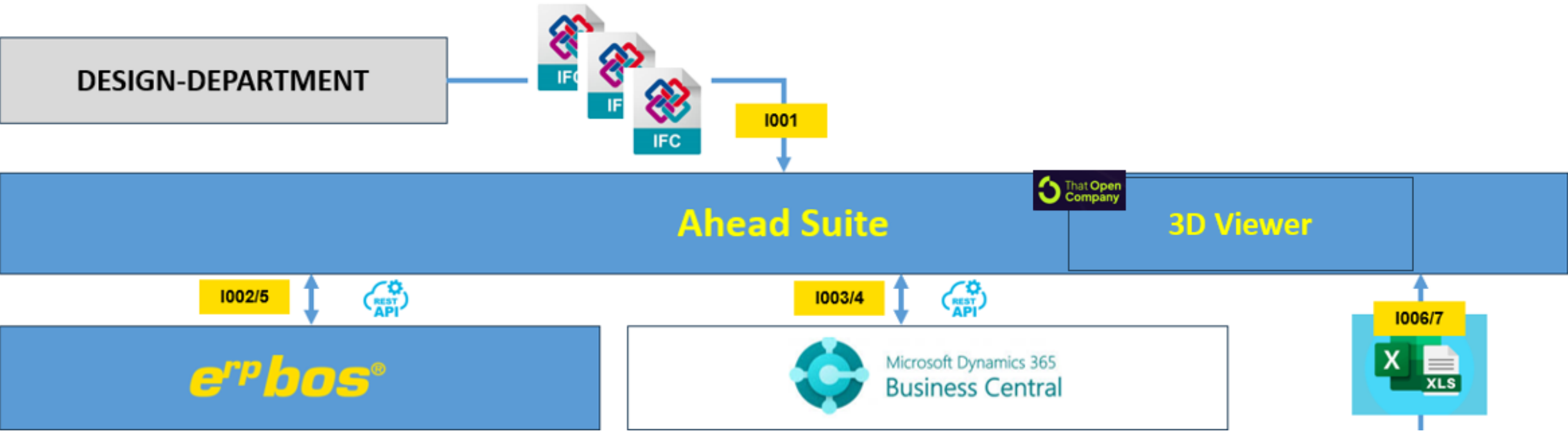
CONNECTING BIM AND ERP

A Unified Planning Platform for Large Construction Projects

PROJECT BACKGROUND

- Customer looked for a central planning platform to manage large construction projects
- Multiple sub-companies involved in these projects using different ERP systems





- Open standard file format for exchanging BIM data between different software tools
- Stores building information such as geometry, materials, spaces, and technical properties
- Parameterization is not fully standardized and can vary between projects, companies and software tools

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```



About ▾ Standards ▾ Compliance ▾ User ▾ Community ▾ Awards ▾ Resources ▾ Events ▾ 

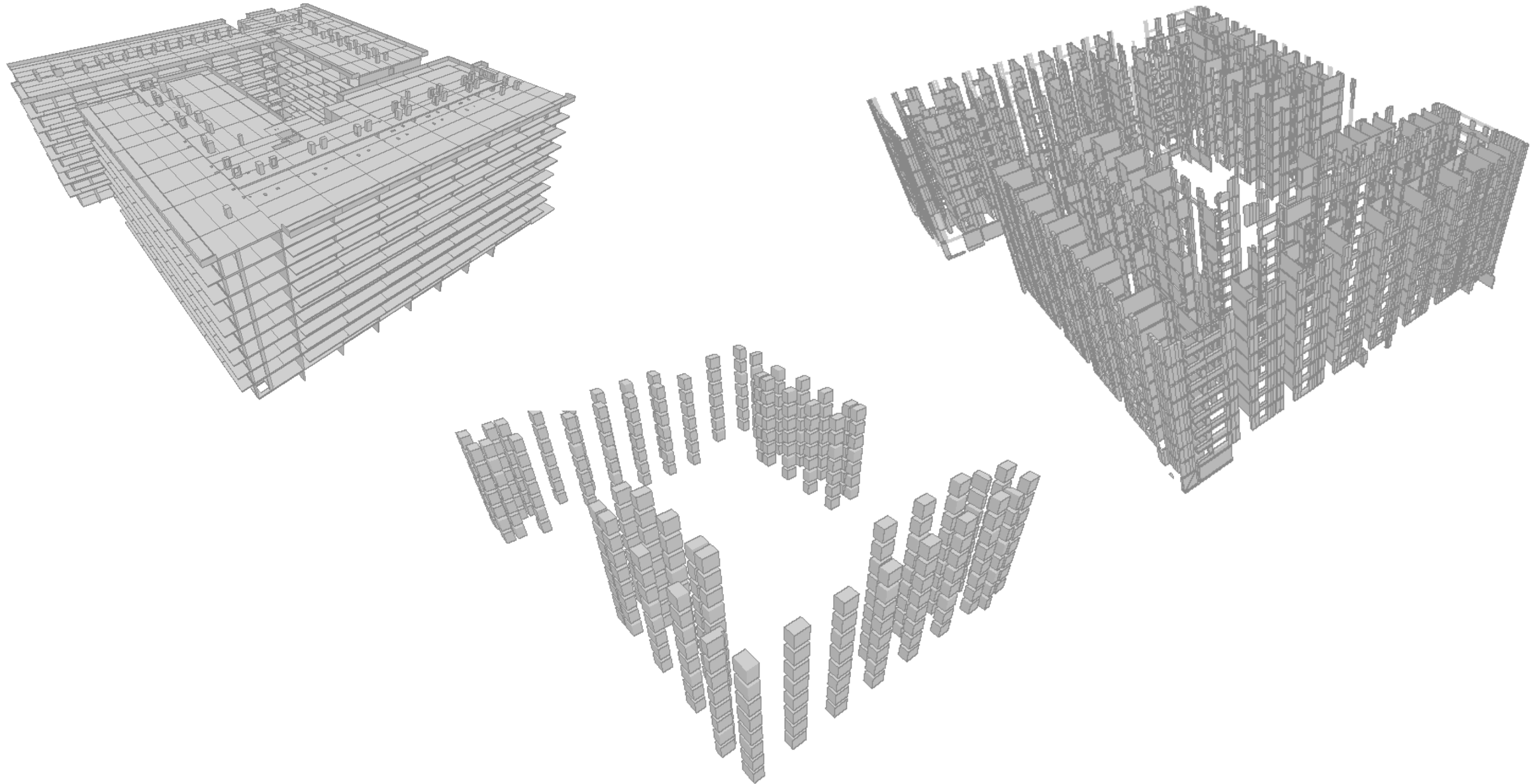
Industry Foundation Classes (IFC)

Industry Foundation Classes (IFC) are a set of standardized, digital descriptions of the built asset industry. It is an open, global standard published under a Creative Commons license, and as ISO 16739.

IFC provides machine interpretability of information and thereby enables automation of workflows. It is vendor-neutral and available to everyone.



DIFFERENT IFC MODELS FOR DIFFERENT PRODUCTS



- IFC Import rules must be flexible as each customer and project require different data mappings
- Rules can be configured by using IFC properties, fixed values or C# code

IFC import rules

Name	Standard
Default	☒

IFC import rule: Default

Filter rules

Type	Operator	Property set	Property	Value
Code	Equals			(ProductWrapper.GetPropertySet("Ba
Code	Equals			(FileNameInfo.Specialization == "ST"
Code	Equals			((FileNameInfo.ProductGroup == "AA
Property	Equals		ProductWrapper.Product.ExpressType	IFCWINDOW
Property	Equals		ProductWrapper.Product.ExpressType	IFCDOOR
Property	Equals		ProductWrapper.Product.ExpressType	IFCFURNITURE
Property	Equals		ProductWrapper.Product.ExpressType	IFCRAILING
Property	Equals		ProductWrapper.Product.ExpressType	IFCTRANSPORTELEMENT
Property	Equals		ProductWrapper.Product.ExpressType	IFCPIPESEGMENT
Property	Equals		ProductWrapper.Product.ExpressType	IFCBUILDINGELEMENTPROXY
Property	Equals		BC_Elem típus	Symbol Fixture
Property	Equals		BC_Elem típus	födémátörés
Property	Equals		BC_Elem típus	Wall niche

Field mappings

Field	Type	Action	Value	Property set	Property
Material	Code	Update	FileNameInfo.ProductGroup !=		
Material	Code	Create	ProductWrapper.GetPropertyV		
Element Name	Code	Create & Update	ProductWrapper.GetPropertyV		
Item Name	Code	Create & Update	ProductWrapper.GetPropertyV		
Quantity	Code	Create & Update	np(IFCLine.ModElementName		
Product Group	Code	Create & Update	FileNameInfo.Specialization ==		
Item Key	Fixed	Create & Update	ITEM-UNIT		
Weight	Property	Create & Update			BC_Suly
Net Width	Property	Create & Update			BC_Netto szelesseg
Height	Property	Create & Update			BC_Magassag
Gross Width	Property	Create & Update			BC_Netto szelesseg
Net Length	Property	Create & Update			BC_Netto hossz
Volume	Property	Create & Update			BC_Terfogat

- Configuration supported by C# coding
- Code validation

Edit

Type

Code

Action

Create & Update

Field *

Element Name

Value *

ProductWrapper.GetPropertyValue("CAD property set", "Elem") ??
ProductWrapper.Name ?? string.Empty

```
internal class FilterRule
{
    public FilterRule(IfcImportFilterRule filterRule)
    {
        var config = new ParsingConfig();
        config.UseDefaultDynamicLinqCustomTypeProvider([typeof(Regex), typeof(RegexOptions)]);

        // convert property filters to code and precompile it
        if (filterRule.Type == IfcFilterType.Code)
        {
            Code = DynamicExpressionParser.ParseLambda<ProductFilterObject, bool>(config, false, filterRule.Value).Compile();
        }
    }
}
```

```
foreach (var filterRule in filterRules)
{
    productFilterObjects =[.. productFilterObjects.Where(filterRule.Code)];
}
```

Edit

Type

Code

Action

Create & Update

Field *

Element Name

Value *

ProductWrapper.GetPropertyValue("CAD property set", "Elem") ?d?
ProductWrapper.Name ?? string.Empty

Apply

Close

	Type	Action	Value	Property set	Property
☐	Material	Code	Update	FileNameInfo.ProductGroup !=	
☐	Material	Code	Create	ProductWrapper.GetPropertyValue	
☐	Element Name	Code	Create & Update	ProductWrapper.GetPropertyValue	
☐	Item Name	Code	Create & Update	ProductWrapper.GetPropertyValue	
☐	Quantity	Code	Create & Update	np(IFCLine.ModElementName	
☐	Product Group	Code	Create & Update	FileNameInfo.Specialization ==	
☐	Item Key	Fixed	Create & Update	ITEM-UNIT	
☐	Weight	Property	Create & Update		BC_Suly
☐	Net Width	Property	Create & Update		BC_Netto szelesseg
☐	Height	Property	Create & Update		BC_Magassag
☐	Gross Width	Property	Create & Update		BC_Netto szelesseg
☐	Net Length	Property	Create & Update		
☐	Volume	Property	Create & Update		

No property or field 'd' exists in type 'ProductFilterObject'

IMPORT AND 3D VISUALIZATION

PROGRESS GROUP

IFC File



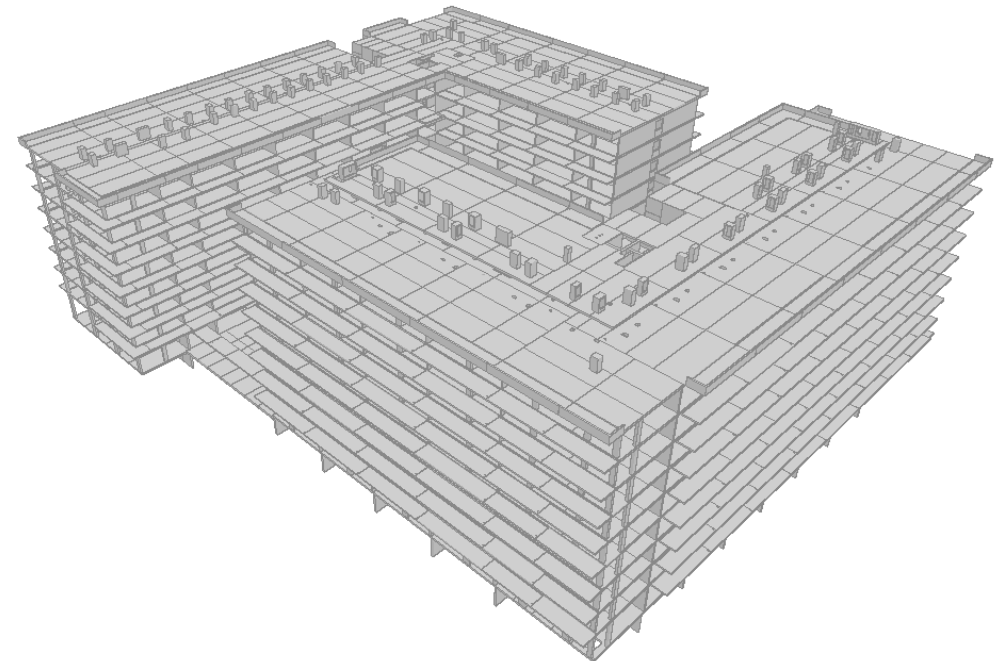
Optimized IFC File

Fragments converter

Web-IFC Viewer

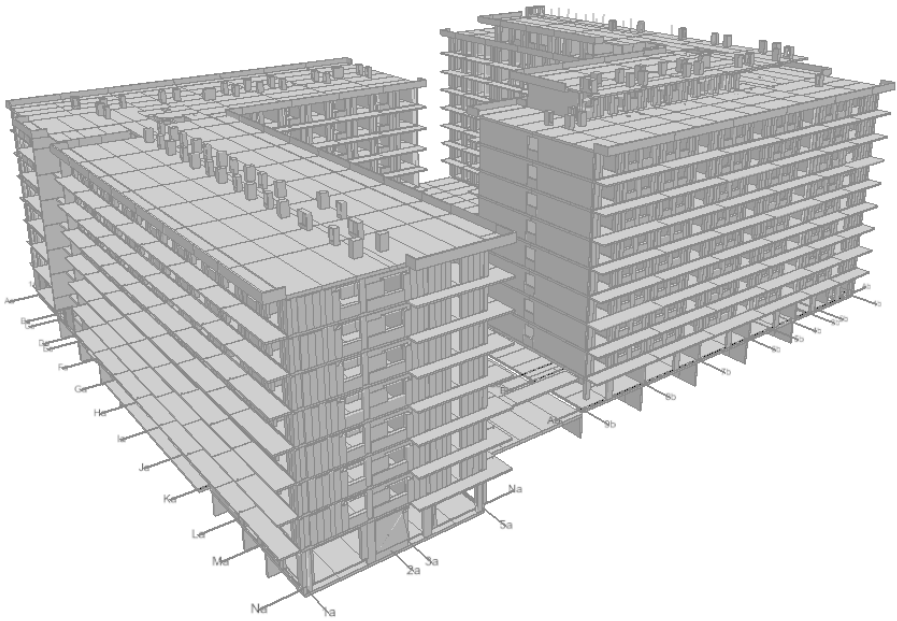


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```



[illegible]

GRIDS AND PERSPECTIVES FOR ORIENTATION



Perspective ▾

Top view

Fit to frame

Multiselect

Details

Filter

Colorize

Edit colors

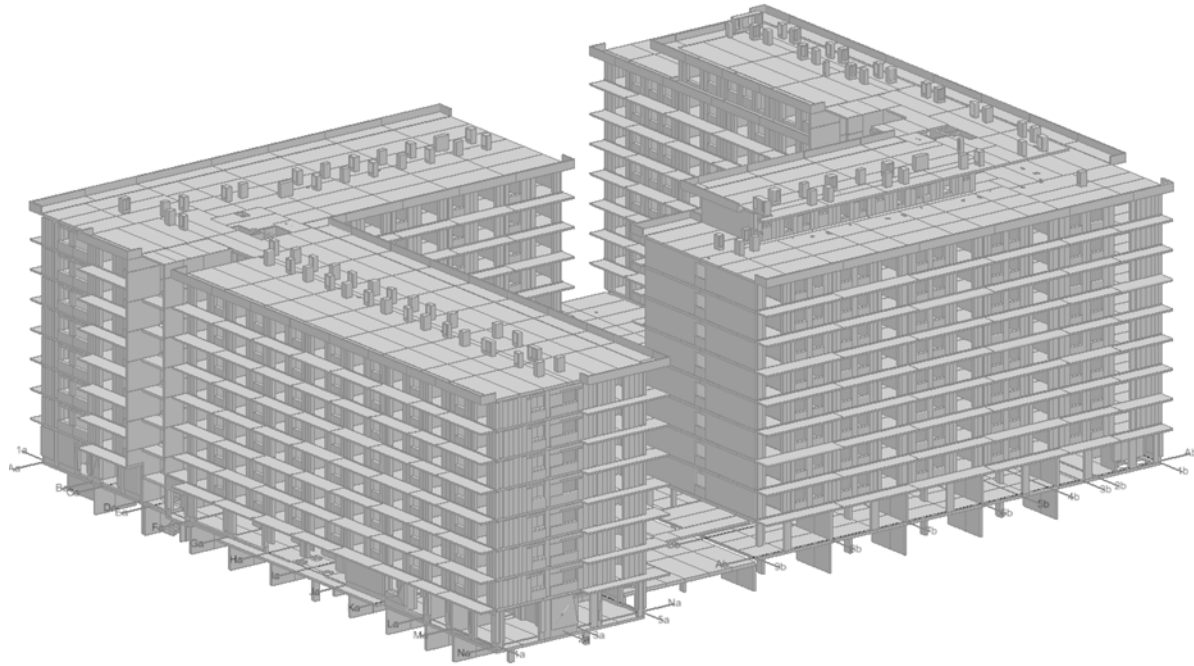
Transport group selection

✂

🛒

🗑

#



Orthographic ▾

Top view

Fit to frame

Multiselect

Details

Filter

Colorize

Edit colors

Transport group selection

✂

🛒

🗑

#

PROGRESS GROUP

aheadSuite

BIM Management

Project: KINCS

BIM VIEWER | TIMELINE | FILES | SETTINGS | DOCUMENTS

Building: K6 Product Group: STR Filter (2) Refresh

ELEMENTS BASKET

293 / 5090 (32)

	ERP Weight	Floor	Room Number	Product
☐	quals	08 emelet	Contains	DD/lv
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	9,732.6 kg	08 emelet	01/01/1	
☐	9,028.7 kg	08 emelet	01/01/1	
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☐	0 kg	08 emelet		
☒	7,916.1 kg	08 emelet	01/01/1	
☒	7,442.6 kg	08 emelet	01/01/1	
☐	7,995.1 kg	08 emelet	01/01/1	
☐	6,733.6 kg	08 emelet	01/01/1	
☒	4,538.7 kg	08 emelet	01/01/1	
☒	4,542.4 kg	08 emelet	01/01/1	
☐	5,270.8 kg	08 emelet	01/01/1	

Update Requested Delivery Date

Requested Delivery Date *

December 2026 < >

M	T	W	T	F	S	S
		1	2	3	4	5
		6				
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Close Save

Project: KINCS

BIM VIEWER

TIMELINE

FILES

SETTINGS

⌵

⌶

🔍

🔍

Building	Start	End	May 2026							
🔍	🔍 📅	🔍 📅	- Sun, 3 May	Mon, 4 May - Sun, 10 May	Mon, 11 May - Sun, 17 May	Mon, 18 May - Sun, 24 May	Mon, 25 May - Sun, 31 May	Mon, 1 Jun - Sun, 7 Jun	Mon, 8 Jun - Sun, 14 Jun	
▼ KINCS	11/02/2026	31/12/2026								
▼ K6	05/05/2026	31/12/2026		K6						
▸ 00 Földszint	31/12/2026	31/12/2026								
▸ 01 emelet	31/12/2026	31/12/2026								
▸ 02 emelet	Update requested delivery date 026	31/12/2026		02 emelet						
▸ 03 emelet	05/05/2026	31/12/2026		03 emelet						
▸ 04 emelet	05/05/2026	31/12/2026		04 emelet						
▸ 05 emelet	05/05/2026	31/12/2026		05 emelet						
▸ 06 emelet	05/05/2026	31/12/2026		06 emelet						
▸ 07 emelet	05/05/2026	31/12/2026		07 emelet						
▸ 08 emelet	05/05/2026	31/12/2026		08 emelet						
▸ 09 emelet	05/05/2026	31/12/2026		09 emelet						

[Purchase Orders](#)

Click to select filters

Filter (0)

275

☐	Purchase Order No.	Supplier No.	Supplier Name	Doc. State	Document Date	Required Date	Project	Int. State	Buildings	Modified on ↓	
☐	PUR-ORD-2026-00180	ERPBOS	ERPBOS	Submitted	22/04/2026	31/12/2026	KINCS	To Receive and Bill	K6	24/05/2026, 22:34:31	→
☐	PUR-ORD-2026-00195	BC - BAYER	BC - BAYER	Draft	15/05/2026	15/05/2026		Draft		24/05/2026, 22:34:19	→
☐	PUR-ORD-2026-00197	ERPBOS	ERPBOS	Submitted	24/05/2026	30/09/2026	KINCS	To Receive and Bill	K6	24/05/2026, 22:33:20	→
☐	PUR-ORD-2026-00192	ERPBOS	ERPBOS	Submitted	15/05/2026	31/12/2026	KINCS	To Receive and Bill	K7	19/05/2026, 14:25:42	→
☐	PUR-ORD-2026-00196	ERPBOS	ERPBOS	Submitted	19/05/2026	29/12/2026	KINCS	To Receive and Bill	K7	19/05/2026, 14:15:37	→
☐	PUR-ORD-2026-00193	ERPBOS	ERPBOS	Submitted	15/05/2026	30/12/2026	KINCS	To Receive and Bill	K1	16/05/2026, 20:44:52	→
☐	PUR-ORD-2026-00190	ERPBOS	ERPBOS	Submitted	15/05/2026	29/12/2026	KINCS	To Receive and Bill	K7	16/05/2026, 20:36:08	→
☐	PUR-ORD-2026-00194	ERPBOS	ERPBOS	Submitted	15/05/2026	29/12/2026	KINCS	To Receive and Bill	K1	15/05/2026, 15:18:07	→
☐	PUR-ORD-2026-00191	BC - VIASTEIN	BC - VIASTEIN	Draft	15/05/2026	29/12/2026	KINCS	Draft	K7	15/05/2026, 11:57:50	→
☐	PUR-ORD-2026-00189	ERPBOS	ERPBOS	Draft	13/05/2026	31/12/2026	KINCS	Draft	K6	13/05/2026, 18:37:55	→
☐	PUR-ORD-2026-00188	ERPBOS	ERPBOS	Draft	11/05/2026	31/12/2026	KINCS	Draft	K6	13/05/2026, 14:32:44	→
☐	PUR-ORD-2026-00186	ERPBOS	ERPBOS	Cancelled	06/05/2026	29/12/2026	KINCS	Cancelled	K6	11/05/2026, 14:25:17	→
☐	PUR-ORD-2026-00187	ERPBOS	ERPBOS	Submitted	08/05/2026	30/12/2026	KINCS	To Receive and Bill	K6	08/05/2026, 18:36:19	→
☐	PUR-ORD-2026-00185	ERPBOS	ERPBOS	Submitted	05/05/2026	30/12/2026	KINCS	To Receive and Bill	K6	05/05/2026, 19:30:51	→
☐	PUR-ORD-2026-00184	ERPBOS	ERPBOS	Submitted	05/05/2026	30/12/2026	KINCS	To Receive and Bill	K6	05/05/2026, 19:28:08	→
☐	PUR-ORD-2026-00181	ERPBOS	ERPBOS	Submitted	27/04/2026	30/12/2026	KINCS	To Receive and Bill	K6	05/05/2026, 19:26:05	→
☐	PUR-ORD-2026-00183	ERPBOS	ERPBOS	Draft	05/05/2026	30/11/2026	KINCS	Draft	K6	05/05/2026, 19:00:22	→
☐	PUR-ORD-2026-00182	ERPBOS	ERPBOS	Draft	29/04/2026	30/04/2026	KINCS	Draft	K6	29/04/2026, 15:08:25	→
☐	PUR-ORD-2026-00179	ERPBOS	ERPBOS	Draft	01/04/2026	30/12/2026	KINCS	Draft	K6	01/04/2026, 18:53:41	→
☐	PUR-ORD-2026-00178	ERPBOS	ERPBOS	Draft	27/03/2026	28/03/2026	KINCS	Draft	K6	27/03/2026, 16:12:15	→
☐	PUR-ORD-2026-00172	ERPBOS	ERPBOS	Draft	13/03/2026	31/12/2026	KINCS	Draft	K6	24/03/2026, 13:36:00	→
☐	PUR-ORD-2026-00177	ERPBOS	ERPBOS	Draft	20/03/2026	30/12/2026	KINCS	Draft	K6	20/03/2026, 11:54:34	→
☐	PUR-ORD-2026-00079	BC - VIASTEIN	BC - VIASTEIN	Submitted	23/01/2026	24/01/2026	KINCS	To Receive and Bill	K6	18/03/2026, 14:42:41	→
☐	PUR-ORD-2026-00176	ERPBOS	ERPBOS	Cancelled	17/03/2026	26/12/2026	KINCS	Cancelled	K6	17/03/2026, 15:25:52	→
☐	PUR-ORD-2026-00175	ERPBOS	ERPBOS	Submitted	17/03/2026	29/12/2026	KINCS	To Receive and Bill	K6	17/03/2026, 15:24:43	→
☐	PUR-ORD-2026-00174	ERPBOS	ERPBOS	Cancelled	17/03/2026	21/12/2026	KINCS	Cancelled	K6	17/03/2026, 15:24:20	→
☐	PUR-ORD-2026-00173	ERPBOS	ERPBOS	Draft	16/03/2026	24/12/2026	KINCS	Draft	K6	16/03/2026, 14:18:04	→

UPDATE ASSEMBLY STATE

aheadSuite

BIM Management

Project: KINCS

BIM VIEWER

TIMELINE

FILES

SETTINGS

Building: K6

Floor: 00 Földszint

Filter (2)

Update Requested Delivery Date

Update Planned Drawing Date

Update IPC State

Update Assembly State

Create Transport Group

Update properties

ELEMENTS

BASKET

753 (40)

	Element Type	Element Name	Floor	Req. Delivery Dat	
☐	gerenda	XY-RB022-00001	00 Földszint	30/12/2026	
☐	gerenda	XY-RB022-00001	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00088	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00011	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00001	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00077	00 Földszint	30/12/2026	
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☐	kéregfal	XY-TW030-00004	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00072	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00084	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00073	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00022	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00056	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00021	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00054	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00078	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00082	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00065	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00003	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00086	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00006	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00076	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00057	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW025-00011	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00023	00 Földszint	30/12/2026	
☐	kéregfal	XY-TW030-00035	00 Földszint	30/12/2026	

Update Assembly State

Assembly State *

Erected on Site

Assembly Date *

05/05/2026

Serial No

Close

Save

Perspective

Top view

Fit to frame

Multiselect

Details

Filter

Colorize

Edit colors

Transport group selection

CHALLENGE: AVOIDING DATA COLLISIONS

- Data can be updated simultaneously by IFC Imports, ERP Updates, manual user input

aheadSuite

Background Jobs

Background Jobs

Text Filter

Show only enabled Jobs ☐

8

☐	Name	Class ↓	Cron-Expression
☐	ERPPOS - Get Delivery Order Updates	UpdateDeliveryOrdersERPbos	*/* * * * *
☐	Dalux Sync	SyncIfcFilesBackgroundJob	*/* * * * *
☐	BC AAC - Element Details	Synchronize MS BC element details	10/15 * * * *
☐	BC BATH - Element Details	Synchronize MS BC bathroom element details	5/15 * * * *
☐	ERPPOS Element Details	Synchronise ERPbos element details	0/17 * * * *
☐	IFC Import	IfcImportBackgroundJob	*/* * * * *
☐	Data Export to Azure	IFCAzureSyncBackgroundJob	0 7/3 * * *

Project: KINCS

BIM VIEWER | TIMELINE | FILES | SETTINGS

Building: K7

Filter (1)

Update Requested Delivery Date

Update Planned Drawing Date

Update IPC State

Update Assembly State

Create Transport Group

Update properties

Update Assembly State

Assembly State *

Erected on Site

Assembly Date *

05/05/2026

Serial No

Close

Save

CHALLENGE: AVOIDING DATA COLLISIONS

PROGRESS GROUP

- Generic update query builder for EntityFramework with **field-based data collision detection**

```
[Generator]
0 references | Markus Pörmbacher, 11 days ago | 2 authors, 3 changes
public class UpdateModelGenerator : IIncrementalGenerator
{
    0 references | Markus Pörmbacher, 128 days ago | 1 author, 1 change
    public void Initialize(IncrementalGeneratorInitializationContext context)
    {
        // Find all record classes with [GenerateUpdate] attribute
        var provider = context.SyntaxProvider
            .CreateSyntaxProvider(
                predicate: (s, _) => IsSyntaxTargetForGeneration(s),
                transform: (ctx, _) => GetSemanticTargetForGeneration(ctx)
            )
            .Where(m => m is not null)
            .Collect();

        context.RegisterSourceOutput(provider, (spc, source) =>
        {
            foreach (var classInfo in source)
            {
                if (classInfo != null)
                {
                    var code = CreateUpdateModelCode(classInfo);
                    spc.AddSource($"{classInfo.ClassName}Update.g.cs", code);
                }
            }
        });
    }
}
```

```
1 reference | Markus Pörmbacher, 11 days ago | 2 authors, 3 changes
private static string CreateUpdateModelCode(ClassInfo classInfo)
{
    var sb = new StringBuilder();

    sb.AppendLine("#nullable enable");
    sb.AppendLine("using Ahead.BL.DataSources;");
    sb.AppendLine();

    sb.AppendLine($"namespace Ahead.BL.Models;");
    sb.AppendLine();

    sb.AppendLine($"/// <summary>");
    sb.AppendLine($"/// Auto-generated Update Model for {classInfo.ClassName}");
    sb.AppendLine($"/// </summary>");
    sb.AppendLine($"public record {classInfo.ClassName}Update : ExternalEntityUpdate");
    sb.AppendLine("{");

    foreach (var prop in classInfo.Properties)
    {
        sb.AppendLine($"    public Trackable<{prop.Type}>? {prop.Name} {{ get; init; }}");
    }

    sb.AppendLine("}");

    return sb.ToString();
}
```


THE BIGGEST CHALLENGE: PERFORMANCE

- Performance of IFC import, 3D visualization and all kinds of data updates heavily depend on the data amount of the IFC models
- Optimization by:
 - Removing not needed data from the IFC files
 - Preparing (binary) format during the IFC import which can be used by the 3D viewer

Original file	➡	 KINCS-GY-K7-ST-BIM-XX-XX-001-M29.ifc	2026. 04. 01. 11:40	IFC File	76 193 KB
After removing not needed data	➡	 KINCS-GY-K7-ST-BIM-XX-XX-001-M29.ifc	2026. 04. 01. 11:40	IFC File	14 184 KB
Binary transformation	➡	 KINCS-GY-K7-ST-BIM-XX-XX-001-M29.ifc	2026. 04. 01. 11:40	IFC File	678 KB

- IFC Viewer: That Open
 - IFC viewer components
 - Customizing can be done through components and Three.js
 - <https://thatopen.com/>
- IFC Importer: xbim
 - Toolkit to read and write BIM models in the IFC format
 - <https://docs.xbim.net/>



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